Real Data for Information Theory

A list of datasets and questions for the group projects

Information theory course
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Each entry describes a dataset, where to obtain it, and some questions related to information theory. Chapters and sections refer to Cover and Thomas (2006). All links were checked on 7 October 2026.

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1 Chance, games and bets

1.1 Loaded dice

Data. Three historical experiments. W. F. R. Weldon recorded 26 306 throws of twelve dice, counting the fives and sixes in each throw, and Karl Pearson used these data in 1900 to present his chi-square test (R package `vcd`, dataset `WeldonDice`, also at `https://vincentarelbundock.github.io/Rdatasets/csv/vcd/WeldonDice.csv`). Labby (2009) repeated the experiment with a machine and a camera. Rudolf Wolf threw a red and a white die together 20 000 times in the nineteenth century. The counts of Labby and Wolf are in Appendix A.

Questions.

- Were the dice fair? How far, in bits per throw, are the observed frequencies from those of fair dice?

- How many throws are needed to detect a bias as small as the one observed? Was Weldon's experiment long enough?
- Jaynes (1983) argued that Wolf's white die had two physical defects and recovered them from the frequencies with the maximum entropy principle. Can you find them?
- Are the red and the white die independent? Did Labby's machine reproduce Weldon's hands?

1.2 The Mega-Sena lottery

Data. Every draw since March 1996 on the page of Caixa, https://loterias.caixa.gov.br/Paginas/Mega-Sena.aspx. The button "Download de resultados" gives a spreadsheet with the six numbers of each draw and the number of winners and the prize for six, five and four hits. When checked, the spreadsheet ended in January 2026, and later draws were available one at a time from the same site.

Questions.
- Are all numbers equally likely? How many bits of uncertainty does one draw resolve?
- Do players prefer some numbers, for instance those that can be days of a month? What do the numbers of winners reveal about the way people choose?

1.3 Football and the bookmaker

Data. Football-Data, https://www.football-data.co.uk/data.php: free CSV files with results and the odds of several bookmakers for 22 European divisions, results since 1993/94 and odds since 2000/01. Brazilian Série A from 2012 to 2026 with closing odds in https://www.football-data.co.uk/new/BRA.csv. Column names are explained in https://www.football-data.co.uk/notes.txt. The site asks that files be downloaded by hand, for private use.

Questions.
- Odds are probabilities with money behind them. What probabilities do they imply, and how many bits per match does the bookmaker's margin cost a bettor?
- How much information do the odds carry about the result? Do outcomes happen as often as the odds say?
- Can a model built only from past results beat the market? What would betting in proportion to your probabilities, as in the horse race of Chapter 6 of Cover and Thomas, have done to a bankroll?
- Is there a home advantage, how large is it in bits, and did it shrink when matches were played without spectators in 2020?

1.4 Horse races in Hong Kong

Data. Kaggle dataset "Horse Racing in HK", https://www.kaggle.com/datasets/gdaley/hkracing, CC0 (free Kaggle account needed): races and runners with the win odds at the start of each race. Dates are deliberately hidden.

Questions.
- This is the horse race of Cover and Thomas, Chapter 6, with real odds. Do favourites win as often as the odds say?
- Does any information in the data (past results of horses and jockeys, track conditions) raise the growth rate of a bettor above that of the public odds?

1.5 Tennis

Data. Results of men's professional matches since 2000 and women's since 2007, with rankings and the odds of several bookmakers (from 2001), one spreadsheet per season, at http://www.tennis-data.co.uk/alldata.php (for private use). Shot-by-shot records of thousands of matches from the Match Charting Project, https://github.com/JeffSackmann/tennis_MatchChartingProject, CC BY-NC-SA 4.0.

Questions.
- How many bits about the winner do the rankings carry, and how many do the odds? Are women's or men's matches, grass or clay, more predictable?
- A good server should be hard to predict. How much uncertainty is there in the direction of a serve, and does it depend on the previous point?

1.6 Chess

Data. Lichess open database, https://database.lichess.org, CC0: more than eight billion rated games in monthly files, with player ratings (Glicko-2), clock times since 2017, and computer evaluations for a few percent of the games. One month is already large, so a small part is enough.

Questions.
- How many bits about the result does the rating difference carry? How good are rating-based predictions?
- How varied are the opening moves at different rating levels?

- How much can a game be compressed? Do strong players make more predictable moves?

1.7 The hot hand

Data. More than 600 000 free throws of the NBA from 2006 to 2016, with the game, the player and the time of each, Kaggle dataset "NBA Free Throws", https://www.kaggle.com/datasets/sebastianmantey/nba-free-throws. All shots of the 2014–15 season, "NBA shot logs", https://www.kaggle.com/datasets/dansbecker/nba-shot-logs. The license of both is not stated.

Questions.

- Does a player who has just scored have a better chance of scoring again? How many bits about the next shot does the last one carry?
- Gilovich et al. (1985) found no hot hand. Miller and Sanjurjo (2018) showed that the usual way of measuring streaks is biased in short sequences. Where does the bias come from, and does it change the conclusion?

1.8 Counting atoms

Data. Rutherford and Geiger (1910) counted α-particles in 2608 intervals of 7.5 seconds. R package `VGAM`, dataset `ruge`.

Questions.

- Are the counts Poisson? How far from Poisson, in bits?
- How precisely can the rate of decay be estimated from these data, and what is the best precision possible?

2 Language, music and code

2.1 How many bits per letter?

Data. `text8`: 10^8 characters of English Wikipedia reduced to lower-case letters and spaces, and `enwik8`, `enwik9`: the raw text with its markup, at https://mattmahoney.net/dc/textdata.html. Results of hundreds of compressors on `enwik9` are listed at https://mattmahoney.net/dc/text.html. Books in English and Portuguese from Project Gutenberg, https://www.gutenberg.org, for instance Machado de Assis's *Dom Casmurro* (ebook 55752), *Memórias Póstumas de Brás Cubas* (54829) and *Quincas Borba* (55682) in the original spelling. Gutenberg asks that bulk downloads use its mirrors. Brazilian literature is also at https://literaturabrasileira.ufsc.br.

Questions.

- Asking people to guess the next letter, Shannon (1951) estimated between 0.6 and 1.3 bits per letter for English. What do letter frequencies, longer contexts, compressors and a modern language model give? Why do the estimates disagree, and which of them are upper bounds?
- The record of the Hutter Prize for compressing `enwik9` was 100 424 672 bytes in September 2026, decompressor included. How far is `gzip` from it, and why?
- Play Shannon's guessing game, or the gambling game of Cover and Thomas, Section 6.6, inside the group. What do you obtain?
- Is Portuguese more or less predictable than English?

2.2 One message, many languages

Data. The Universal Declaration of Human Rights in more than 500 languages, http://efele.net/udhr/ (also in the Python library NLTK). The Bible in 100 languages aligned verse by verse, https://github.com/christos-c/bible-corpus, CC0. Measured information rates of speech in 17 languages from Coupé et al. (2019), https://github.com/keruiduo/SupplMatInfoRate.

Questions.

- Texts with the same content have very different lengths in different languages. Are they more alike after compression?
- Can languages be grouped into families using only a compressor, by measuring how much better a text compresses after the compressor has read a text in another language (Benedetto et al., 2002)?
- Fast-spoken languages seem to pack less information into each syllable, so that information per second is about the same everywhere (Coupé et al., 2019). Do their data support this?

2.3 Who wrote it?

Data. The 85 Federalist Papers, written in 1787 and 1788 by Alexander Hamilton, James Madison and John Jay under one pseudonym. Twelve are disputed between Hamilton and Madison. Project Gutenberg ebook 18, and R package `HistData` (`Federalist_corpus`, `Federalist_markers`, `Federalist_text`).

Questions.

- Mosteller and Wallace (1963) settled the dispute by counting small words such as *upon*. How many bits of evidence about the author does one occurrence of a word carry, and a whole paper?
- How many words does a text need before its author can be identified with small error?
- Can a compressor attribute authorship, by checking which author's texts help most to compress a disputed paper?
- Is there an authorship question of the same kind among Brazilian texts in the public domain?

2.4 Spam or not?

Data. SMS Spam Collection: 5574 text messages in English, 747 of them spam, CC BY 4.0, `https://archive.ics.uci.edu/dataset/228/sms+spam+collection` (Almeida et al., 2011).

Questions.

- How many bits of evidence for spam does each word carry? Which words carry the most?
- Can a compressor recognize spam, by checking which class of messages helps most to compress a new one? How does it compare with a model of word counts?
- How many words of a message are needed for a reliable decision?

2.5 First names

Data. All first names given to at least five babies of each sex in each year from 1880 to 2025 in the United States, `https://www.ssa.gov/oact/babynames/limits.html`. Brazilian first names by decade of birth, state and municipality from the 2010 census, `https://servicodados.ibge.gov.br/api/docs/nomes?versao=2`, and from the 2022 census, with surnames, at `https://censo2022.ibge.gov.br/nomes`. Rare names are omitted.

Questions.

- Has the diversity of first names grown over the decades, in Brazil and in the United States?
- How many bits about sex, decade of birth or state does a first name carry? How well can sex be guessed from the name alone?
- What does the omission of rare names do to these numbers?

2.6 Source code

Data. Any large open-source project, downloaded with `git`, for instance the Python interpreter, `https://github.com/python/cpython`, or the Linux kernel, `https://github.com/torvalds/linux`.

Questions.

- Hindle et al. (2012) found that source code is much more predictable than English text. How many bits per character does code need, compared with prose? Which programming languages are the most predictable?
- How much of that predictability belongs to one project and how much to the language?

2.7 Music

Data. The corpus of the Python package `music21` (Cuthbert and Ariza, 2010): the 371 Bach chorales, 8514 folk songs of the Essen collection, Palestrina masses, Monteverdi madrigals and more, in musical notation (some files for non-commercial use only). The Lakh MIDI Dataset, 176 581 MIDI files of popular music, CC BY 4.0, `https://colinraffel.com/projects/lmd/` (Raffel, 2016). MAESTRO, about 200 hours of piano performances with the notes played, CC BY-NC-SA 4.0, `https://magenta.tensorflow.org/datasets/maestro`.

Questions.

- How many bits per note does a melody need, given the notes before it? Is Bach more predictable than folk songs, or than Palestrina?
- Can a compressor, or a simple model of note sequences, tell composers and styles apart?
- How does surprise, measured in bits, vary along a piece? Do pianists play surprising notes differently?

3 Society and politics

3.1 Births and the clock

Data. SINASC, the national record of live births, one CSV file per year from 1996 to 2024 on the federal open data portal, `https://dados.gov.br/dados/conjuntos-dados/sistema-de-informacao-sobre-nascidos-vivos-sinasc-1996-a-20201`. Each record has the type of delivery, the date and time of birth, the mother's age and schooling, gestational age, birth weight, municipality and hospital code. In 2024 there were 2 389 325 births, 60.6% of them by cesarean section. The R package `microdatasus` and the Python package `PySUS` read the files. The records are de-identified data about people.

Questions.

- How many bits about the type of delivery do the hour and the day of the week of birth carry?
- How concentrated in time are vaginal and cesarean births? What happens on holidays and during Carnival?
- Does the answer depend on the mother's schooling, the region, or the hospital, and how has it changed since the 1990s?
- How precise are the recorded times? Do they pile up on round hours?

3.2 Party discipline

Data. Every recorded vote of every federal deputy since 2001, one file per year, for instance `https://dadosabertos.camara.leg.br/arquivos/votacoesVotos/csv/votacoesVotos-2023.csv`, with party and state, and the vote recommended by each party leader (documentation at `https://dadosabertos.camara.leg.br/swagger/arquivos.html`). Senate votes through `https://legis.senado.leg.br/dadosabertos/votacao`. Every roll call of the US Congress since 1789 at `https://voteview.com/data`.

Questions.

- How many bits about a deputy's vote does the party carry? Has this changed since 2001? How does Brazil compare with the US Congress?
- Does the party carry information beyond the leader's recommendation?
- Which deputies vote alike? Can the parties be recovered from the votes alone?
- Small parties have few deputies. How does that distort the estimates?

3.3 The geography of the vote

Data. Results of the 2022 elections by polling station and by municipality from the electoral court, `https://dadosabertos.tse.jus.br/dataset/resultados-2022`, CC BY. The presidential file by polling station has 1.6 GB.

Questions.

- How many bits about a vote does the voter's state carry? And the municipality, once the state is known? And the polling station, once the municipality is known?
- Which states are the most atypical?
- What is the effective number of candidates in the first round, in the country and in each state?
- Do vote counts follow Benford's law of first digits, and what would a deviation mean (Deckert et al., 2011)?

3.4 ENEM

Data. Microdata of the national secondary-school examination for 1998 to 2025, `https://www.gov.br/inep/pt-br/acesso-a-informacao/dados-abertos/microdados/enem`: each candidate's answers, scores and socioeconomic questionnaire. From 2024 on, questionnaires and scores are in separate files that cannot be linked.

Questions.

- How many bits about the score do family income, parents' schooling or type of school carry?
- How much information does each test item carry, and are some items redundant?
- Do the answer patterns reveal guessing?

3.5 Income in the census

Data. UCI Adult: 48 842 people from the 1994 US census with 14 attributes and whether income exceeds 50 000 dollars, CC BY 4.0, `https://archive.ics.uci.edu/dataset/2/adult`. Its modern successor is the Python package `folktables` (Ding et al., 2021).

Questions.

- Which attributes carry most information about income, and which are redundant given the others?
- What is the smallest error that any classifier could reach with these attributes?
- How much information about sex or race leaks into a classifier's predictions?

4 The planet and the sky

4.1 Earthquakes

Data. The catalogue of the U.S. Geological Survey, public domain, through `https://earthquake.usgs.gov/fdsnws/event/1/`. A web address such as `https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv&starttime=2000-01-01&endtime=2001-01-01&minmagnitude=4.5` returns a CSV file with time, position, depth and magnitude of every event. At most 20 000 events come per request.

Questions.

- The Gutenberg–Richter law says that larger earthquakes are exponentially rarer, and the exponential is the maximum entropy law for a given mean. Is the mean magnitude all there is to know, in Chile, Japan, California and worldwide?
- How precisely can the slope of the law be estimated, and how many earthquakes are needed to detect a change of slope before a large event?
- Does the magnitude of an earthquake carry information about the next one, or about the time until it?

4.2 Rain

Data. Hourly data of the automatic weather stations of INMET, one zip file per year from 2000 to 2026, `https://portal.inmet.gov.br/dadoshistoricos` (São Paulo has stations A701 and A771). Longer daily series of conventional stations can be requested at `https://bdmep.inmet.gov.br/` with an e-mail address. NOAA's GHCN-Daily has per-station files, `https://www.ncei.noaa.gov/data/global-historical-climatology-network-daily/access/`, including São Paulo from 1936.

Questions.
- How much does today's weather tell about tomorrow's? How many past days matter?
- Different criteria for choosing the memory of a model disagree on these data. Why?
- How much do the season and a neighbouring station add?
- Has the pattern of wet and dry days changed between decades?

4.3 Sunspots

Data. Monthly mean sunspot number since January 1749 from the Royal Observatory of Belgium, `https://www.sidc.be/SILSO/datafiles`, license CC BY-NC 4.0.

Questions.
- How predictable is the solar cycle? How many bits does each new month bring once the past is known?
- What does a spectrum estimated by the maximum entropy method of Burg show about the solar cycle?
- Is the dependence between months linear, or is there more information than the correlations reveal?

4.4 A tropical forest

Data. Every tree above a minimum size in a 50-hectare plot of Barro Colorado Island, Panama, with species, size and position, in eight censuses from 1982 to 2015, CC0, `https://doi.org/10.15146/5xcp-0d46`. A small version is the dataset `BCI` of the R package `vegan`.

Questions.
- How diverse is the forest? Ecologists' diversity indices are entropies under other names. How can one account for species not seen in a sample?
- How many bits about the species of a tree does its position carry? Has diversity changed in 35 years?

5 Life

5.1 Genomes

Data. The genome of the bacterium *Escherichia coli* K-12 (4 641 652 bases), `https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/`, with the gene annotation (`genomic.gff.gz`) in the same folder. Human chromosome 21 (46.7 million positions), `https://hgdownload.soe.ucsc.edu/goldenPath/hg38/chromosomes/chr21.fa.gz`.

Questions.
- Two bits per base always suffice. How much less does a genome need? Do ordinary compressors beat two bits per base?
- How far along the sequence does the dependence reach? Why does a period of three appear inside genes?
- How many bases are needed to tell a gene from non-coding DNA?

5.2 Binding sites

Data. JASPAR 2026, `https://jaspar.elixir.no/downloads/`, CC BY 4.0: 2633 non-redundant profiles of the DNA sites bound by transcription factors, given as frequency matrices.

Questions.
- How many bits does a binding site carry? Draw sequence logos (Schneider and Stephens, 1990).
- How many sites would appear by chance in a genome? Schneider et al. (1986) found that the information in a site is close to the information needed to find it in the genome. Is it?

5.3 Ancestry in DNA

Data. The 1000 Genomes Project (The 1000 Genomes Project Consortium, 2015): genomes of 2504 people from 26 populations on five continents, with the frequency of each variant in each continental group, `https://www.internationalgenome.org/data`. Files per chromosome at `https://ftp.1000genomes.ebi.ac.uk/vol1/ftp/release/20130502/` (chromosome 22 has about 200 MB).

Questions.

- How many bits about a person's continent of origin does one genetic marker carry (Rosenberg et al., 2003)? Which markers carry the most?
- How many markers are needed to identify the continent with small error? And the population within a continent?

5.4 Protein families

Data. Pfam alignments of protein families, CC0, served by InterPro: for instance `https://www.ebi.ac.uk/interpro/api/entry/pfam/PF00071/?annotation=alignment:seed`. Bulk files at `https://ftp.ebi.ac.uk/pub/databases/Pfam/current_release/`.

Questions.

- Which pairs of positions of a protein vary together across species? Does their mutual information reveal which amino acids touch in the folded protein (Dunn et al., 2008)?

5.5 A signalling network

Data. Eleven signalling molecules measured in thousands of single cells under several perturbations (Sachs et al., 2005), with a known network: `https://www.bnlearn.com/research/sachs05/` and `https://zenodo.org/records/7681811` (CC BY 4.0).

Questions.

- Can the known network be recovered from the information between pairs of molecules? Which links are indirect?

5.6 Neurons

Data. Recordings of neurons in open archives: DANDI, `https://dandiarchive.org` (no account for public data), the Allen Brain Observatory Visual Coding Neuropixels data (Python package `allensdk`), and CRCNS, `https://crcns.org` (free account).

Questions.

- How many bits per spike does a neuron transmit about the stimulus (Strong et al., 1998)? How much data does a reliable estimate need?

5.7 Heartbeats

Data. PhysioNet Fantasia database, `https://physionet.org/content/fantasia/1.0.0/`: two hours of electrocardiogram with annotated beats from 20 young and 20 elderly healthy people. Also the MIT-BIH Arrhythmia Database, `https://physionet.org/content/mitdb/1.0.0/`. Open access.

Questions.

- Is the heartbeat of the elderly more or less predictable than that of the young? How many bits does each beat bring?
- Physiologists use "approximate entropy" and "sample entropy". What do these numbers actually measure?

5.8 Birdsong and whale codas

Data. Songs of four Bengalese finches annotated syllable by syllable, CC BY 4.0, `https://doi.org/10.6084/m9.figshare.4805749` (Nicholson et al., 2017). Phrase sequences of Cassin's vireo songs, CC BY 4.0, `https://doi.org/10.6084/m9.figshare.3081814.v1`, whose annotations take only 2.2 MB (Hedley, 2016). 8719 codas of sperm whales recorded off Dominica, with rhythm, coda type, clan, social unit and whale, `https://github.com/pratyushasharma/sw-combinatoriality` (Sharma et al., 2024), terms of use not stated. Earlier codas with individual identities, CC0, `https://doi.org/10.5061/dryad.ck4h0` (Gero et al., 2016).

Questions.

- Is a birdsong a Markov chain? How many bits does each new syllable add, and how far back does the song remember?
- How many bits about a whale's clan, family unit or identity does one coda carry? Do the rhythms carry more than the coda types?

5.9 Hearing in noise

Data. Miller and Nicely (1955) recorded how listeners confused 16 English consonants under different levels of noise. The tables are in the article. A pooled version is the dataset `Phonemes` of the R package `clue`. Confusions between Morse code signals are the datasets `morse` and `morse2` of the R package `smacof`.

Questions.

- Seen as a noisy channel, how many bits per consonant does speech in noise transmit, and what is the capacity of that channel?
- Which features of a consonant survive noise best?

6 Money, forecasts and tastes

6.1 Investing

Data. Daily and monthly returns of US industry portfolios since July 1926, Kenneth French Data Library, `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`. Daily prices of 36 New York stocks from 1962 to 1984, a classic test set, distributed with the Python package `universal-portfolios`, `https://github.com/Marigold/universal-portfolios`. Daily quotes of every security traded in Brazil since 1986 from B3, `https://www.b3.com.br/pt_br/market-data-e-indices/servicos-de-dados/market-data/historico/mercado-a-vista/cotacoes-historicas/` (prices not adjusted for dividends).

Questions.

- Rebalancing a portfolio to fixed proportions every day can grow faster than every single asset in it. When does this happen in real data?
- How close does Cover's universal portfolio (Cover, 1991), which needs no model of the market, come to the best fixed proportions chosen with hindsight, and how fast?
- Can information known in advance, such as last month's return, raise the growth rate, and by how much at most?

6.2 Forecasts

Data. Probabilities published by FiveThirtyEight for elections and sports, paired with what happened, CC BY 4.0, `https://github.com/fivethirtyeight/checking-our-work-data`.

Questions.

- How many bits of information about the outcomes did the forecasts carry? Were they calibrated, that is, did events announced with probability 70% happen about 70% of the time?
- Which are more predictable, elections or sports?

6.3 Films

Data. MovieLens, `https://grouplens.org/datasets/movielens/`: film ratings given by users of a recommendation site. Versions range from 100 000 ratings by 943 users to 32 million ratings by about 200 000 users of 87 585 films (1995 to 2023). For research use, citing Harper and Konstan (2015).

Questions.

- How predictable are ratings? How many bits about a user's rating of one film do the user's other ratings carry, and how many do the ratings of other users carry?
- Are some users' tastes more varied than others'? Has the variety of films rated changed since 1995?

7 Cities, the web and machines

7.1 How predictable are we?

Data. Check-ins in location-based social networks, from Stanford's SNAP collection: Gowalla (6.4 million check-ins of 196 591 users, 2009–2010), `https://snap.stanford.edu/data/loc-Gowalla.html`, and Brightkite (4.5 million check-ins of 58 228 users), `https://snap.stanford.edu/data/loc-Brightkite.html`. GPS trajectories of 182 people in Beijing (GeoLife), `https://www.microsoft.com/en-us/download/details.aspx?id=52367`, for non-commercial use without redistribution.

Questions.

- From the entropy of people's trajectories and Fano's inequality, Song et al. (2010) concluded that where a person is can be predicted with accuracy of at most about 93%. What do these data say?
- How does the answer change with the size of the map cells and with the way the entropy is estimated? Later studies disagree (Lu et al., 2013; Xu et al., 2019).
- Do simple predictors come close to the limit?

7.2 Bike trips

Data. Every trip of New York's Citi Bike system since 2013, with start and end station and times, as yearly and monthly files at `https://citibikenyc.com/system-data`. The data may be used in non-commercial studies but not redistributed. No open trip data from a Brazilian system was found.

Questions.

- How many bits about the destination of a trip does the starting station carry? And the hour of the day, or the day of the week?
- Are bike trips more or less predictable than the check-ins of the previous entry?

7.3 Wikipedia navigation

Data. Wikipedia clickstream, `https://dumps.wikimedia.org/other/clickstream/`, CC0: for each month since 2017, how many times readers went from one page to another, in 40 languages including Portuguese. Pairs seen ten times or fewer are omitted.

Questions.

- How many bits does a reader's next click need? How much does the current article tell about the next one?
- Do readers of the Portuguese and the English Wikipedia navigate in equally predictable ways?
- How varied are the ways readers arrive at an article (search engines, links, other sites)?

7.4 Images

Data. The 24 Kodak test photographs, `https://r0k.us/graphics/kodak/`. Handwritten digits (MNIST) and clothing items (Fashion-MNIST) on Hugging Face as `ylecun/mnist` and `zalando-datasets/fashion_mnist`. CIFAR-10, 60 000 small colour images, `https://www.cs.toronto.edu/~kriz/cifar.html`.

Questions.

- How many bits does a photograph need for a given quality? How close do JPEG and WebP come to the limits given by rate distortion theory?
- Which components of an image carry most of the bits?
- How far can a digit be compressed before a classifier, or a person, no longer recognizes it?

7.5 Network traffic

Data. CIC-IDS2017, five days of network traffic with labelled attacks, `https://www.unb.ca/cic/datasets/ids-2017.html` (short registration form). Known problems with its labels are described by Engelen et al. (2021).

Questions.

- Does the entropy of addresses and ports change during an attack (Lakhina et al., 2005)? How quickly can the change be detected, and with how many false alarms?

7.6 Is it random?

Data. The NIST Randomness Beacon, `https://beacon.nist.gov`, publishes 512 random bits every minute, with past values available one by one through its programming interface. NIST's tests for random numbers (SP 800-22, `https://csrc.nist.gov/pubs/sp/800/22/r1/upd1/final`) and its estimators of entropy for random sources (SP 800-90B, code at `https://github.com/usnistgov/SP800-90B_EntropyAssessment`). The digits of π, which any group can compute, for instance with the Python package `mpmath`. Sequences typed by people trying to be random, which the group can collect.

Questions.

- Do the beacon's bits, the digits of π and human "random" sequences pass the same tests? Can a compressor shorten any of them?
- The digits of π come from a short program, so their Kolmogorov complexity is small (Cover and Thomas, Chapter 14). Why do statistical tests not notice?
- How many bits of entropy per symbol do people really produce when they try to be random?

8 Where to look for more data

- **General:** Kaggle, `https://www.kaggle.com/datasets` (free account). Hugging Face, `https://huggingface.co/datasets` (more than a million datasets). UCI Machine Learning Repository, `https://archive.ics.uci.edu`. OpenML, `https://www.openml.org`. Google Dataset Search, `https://datasetsearch.research.google.com`. Rdatasets, `https://vincentarelbundock.github.io/Rdatasets/` (more than 3600 classic datasets from R packages as CSV). Awesome Public Datasets, `https://github.com/awesomedata/awesome-public-datasets`.

- **Data from scientific articles:** Zenodo, Dryad, Figshare, Harvard Dataverse.
- **Networks and sequences of events:** SNAP, https://snap.stanford.edu/data/, and Network Repository, https://networkrepository.com.
- **Science:** PhysioNet (physiology), NCBI, Ensembl and UCSC (genomes), NOAA (climate), Copernicus Climate Data Store (free account), CERN Open Data (particle physics).
- **World indicators:** World Bank Open Data and Our World in Data.
- **Brazil:** Portal Brasileiro de Dados Abertos, https://dados.gov.br. Base dos Dados, https://basedosdados.org (cleaned Brazilian public data). IBGE SIDRA, https://sidra.ibge.gov.br. Ipeadata, http://www.ipeadata.gov.br. DATASUS (health), INEP (education), TSE (elections), Câmara and Senado (legislative votes), INMET (weather), B3 (stock market).
  Each dataset has its own license or terms of use, stated on its page.

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A Dice data

k	0	1	2	3	4	5	6	7	8	9	10	11	12
Weldon	185	1149	3265	5475	6114	5194	3067	1331	403	105	14	4	0
Labby	216	1194	3292	5624	6186	5047	2953	1288	406	85	13	2	0

Number of throws of twelve dice, out of 26 306, in which exactly k dice showed five or six.

In Labby's experiment the faces 1 to 6 appeared 53 222, 52 118, 52 465, 52 338, 52 244 and 53 285 times among the 315 672 dice thrown.

	White die						
Red die	1	2	3	4	5	6	Total
1	547	587	500	462	621	690	3407
2	609	655	497	535	651	684	3631
3	514	540	468	438	587	629	3176
4	462	507	414	413	509	611	2916
5	551	562	499	506	658	672	3448
6	563	598	519	487	609	646	3422
Total	3246	3449	2897	2841	3635	3932	20000

Wolf's 20 000 throws of a red and a white die (Jaynes, 1983).